

¹³C Nuclear Spins — Comparison of Three Methods' Results

2021 paper – limitations of Dr. Jung's method, with comparison table

	Manual analysis	2021 method (Dr. Jung's method)	WD method
NV1 Dataset	4 spins (−88, 47) (−5, 35) (+8, 31) (−38, 37)*	Output = 5 high-probability positions on the A probability curve (only says 'a spin is near this A ' — A _⊥ , count, and exact values are not output) ① Region A ∈ [−6, +12] (site of 5 WD spins) ② Region A ∈ [+40, +46] (near WD +39/+48) ③ Peak −20 = WD (−18.8) site ④ Peak +22 = WD (+24.3) site ⑤ Peak −34 = none; matches off-list candidate −32	14 spins — (A , A _⊥) (−87.9,18) (−18.8,16) (−11.5,14) (−5.1,22) (+0.6,17) (+4.0,26) (+8.4,28) (+14.2,22) (+24.3,20) (+39.0,28) (+48.0,25) (+65.6,28) (+91.9,33) (+114.4,24)
NV2 Dataset	3 spins (−340, 290) (+150, 110) (−42, 150)*	— Not evaluable: single-channel, strongly-coupled data is out of scope	10 spins — (A , A _⊥) (−151.2,95) (−58.9,182) (−45.9,50) (−38.9,67) (−12.6,42) (−2.7,37) (+51.7,91) (+55.9,55) (+346.9,261) (+430.7,58)*
Notes	Rough position anchors from visual reading — all 7 rediscovered in the WD list. A _⊥ tends to be overestimated	Gets positions right, but the list is incomplete - Outputs only down to region/peak level Later regression stage fixing count and exact A , A _⊥ is out of scope	Count (BIC, consistent with RJMCMC posterior) · complete with A _⊥ values · 95% CI Reproduces 3 measurements simultaneously, RMSE 0.088–0.176

*A_{||} signs from manual analysis differ by m_s branch convention · *withheld due to possible strong coupling or degeneracy
† BIC (Bayesian Information Criterion) = statistical rule that increases the count only when the fit improvement from adding a spin outweighs the complexity penalty
· 95% CI (confidence interval) = 'interval containing the true value with 95% probability' (error bars), measured from the value spread across 60 residual-bootstrap refits

2021 Method vs Our Method

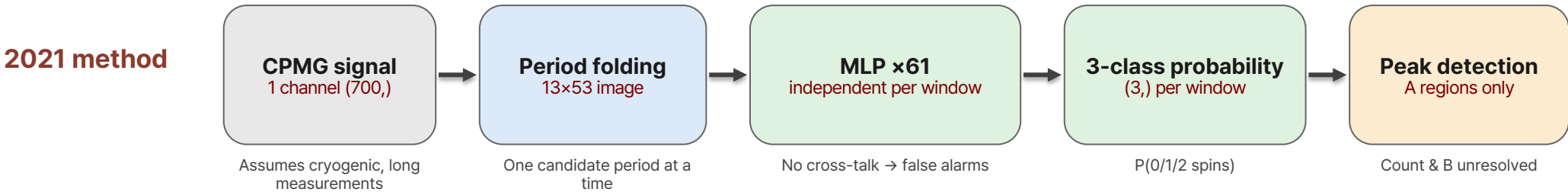
Compared in the same measured spin environment (NV1) — WD method ahead by ×2.2 cryo, ×1.5 room temp, ×1.7 error-injected

	2021 method (core-pipeline reproduction)	WD method (PF→RJMCMC hybrid v2)
Architecture	61 independent per-window MLPs + separate denoiser	Neural detection (cross-window attention) + region-constrained physics fitting + BIC
Output	High-probability regions ($A_{ }$ position candidates); $A_{\perp}$ and count require a separate stage	Final spin list $\{(A_{ }, A_{\perp})\} \times k^*$ — count automatic · 95% CI provided
Measured 50-spin validation (F1)	Cryo 0.38 · room temp 0.37 · error-injected 0.35	Cryo 0.84 · room temp 0.57 · error-injected 0.61
False alarms (precision)	0.55–1.00 (collapses at room temp)	0.87–1.00 (0.87 even with error injection)
Real data NV1	2 high-probability regions + 3 peaks (individual spins not resolvable)	14 spins confirmed (with CI); reproduces 3 measurements simultaneously, RMSE 0.09–0.18
Prerequisites	Cryogenic · high SNR · long $N=32/256$ measurements	Works at room temp · 700 points · $N \leq 20$

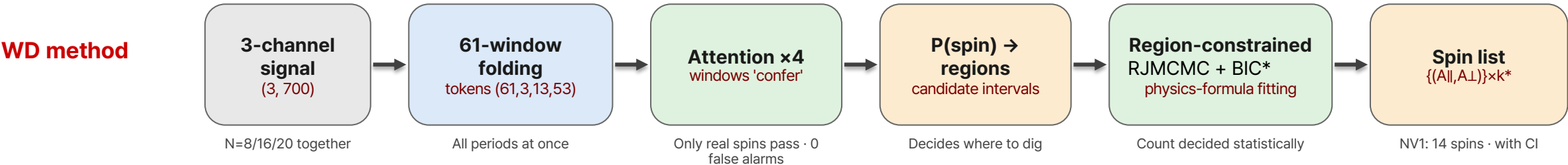
• Fairness — 2021 figures are from a core-pipeline reproduction · both methods use the same forward model and the same scoring rules

Model Comparison: Block Diagram

Only the window-folding step is shared — our model is designed to also determine how many spins there are



⬆ Both start from the same 'period folding' — 2021 stops there (isolated decisions); we bundle the 61 folded windows into tokens that "confer", then focus on the resulting regions and enumerate spins by fitting

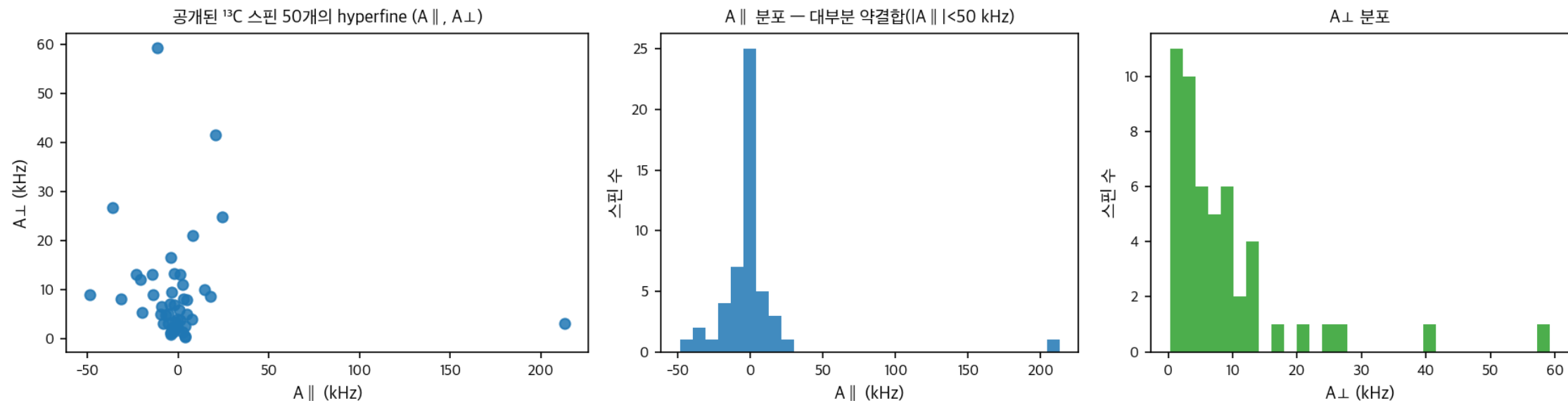


*Reversible-Jump Markov Chain Monte Carlo : https://en.wikipedia.org/wiki/Reversible-jump_Markov_chain_Monte_Carlo
Bayesian information criterion : https://en.wikipedia.org/wiki/Bayesian_information_criterion

Feature	2021 method	WD method
Window decisions	61 windows decide individually (isolated) — no cross-reference → false alarms	Attention lets windows reinforce each other ('confer') → filters out fake signals · 0 false alarms
Spin search	Peak finding without regions — only A position clues	Region-constrained enumeration — physics fitting only within candidate intervals + count decided by BIC
Output	Up to A position candidates (regions/peaks)	Complete spin list: count k^* · A · coupling strength $A_{\perp}$ · 95% CI
Validation (measured 50-spin F1)	0.35 ~ 0.38	0.57 ~ 0.84

Overview of the Public 50-Spin Dataset (van de Stolpe et al. 2024)

A network of 50 ¹³C nuclear spins around a single NV center in diamond — a measured dataset with all per-spin parameters published

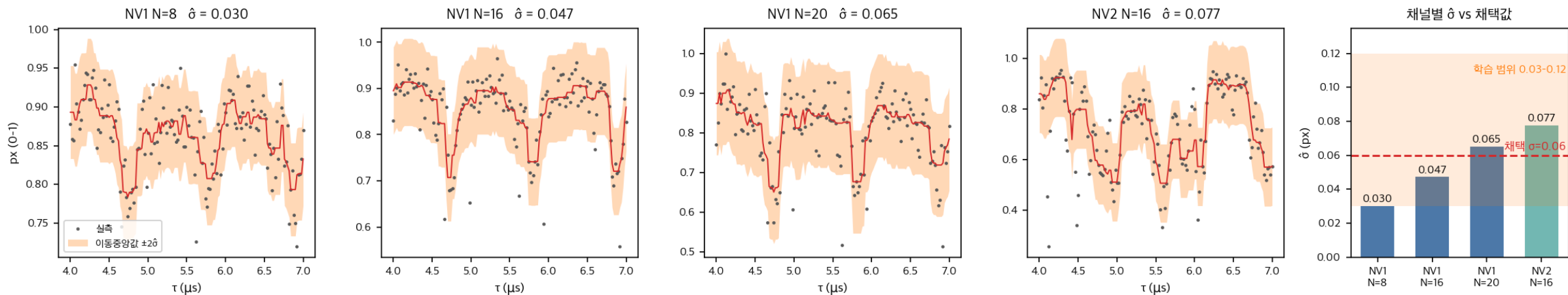


System	One single NV center in diamond + 50 surrounding ¹³ C nuclear spins (+1 nitrogen) — entire network mapped with correlated-sensing (concatenated double-resonance) sequences
Measurement conditions	Cryogenic 3.7 K · B = 403.553 G (along NV axis) · presumably the same NV sample as Jung 2021
Published spin information	Per-spin hyperfine (A : −48.5 to +213.2 kHz, A _⊥ : 0.2 to 59.2 kHz) · spin frequencies · 3D lattice positions · 474 measured nuclear–nuclear couplings (mostly a few Hz)
File contents	spin_table / spin_frequencies / spin_positions / spin_measured(predicted)_couplings (xlsx·pkl) + notebooks reproducing the paper's figures

Noise Modeling — Basis for the Synthetic σ

σ extracted by statistical estimation from measured NV1/NV2 data; scheme for mixing cryogenic data into room-temperature data

Step	Method / Value	Meaning
① σ estimator	Adjacent-point differences $d_i = y_{i+1} - y_i \rightarrow \hat{\sigma} = \text{std}(d)/\sqrt{2}$	Slow signal components cancel out, leaving only noise (standard estimator under an independent-Gaussian assumption)
② Measured results	NV1 $\hat{\sigma} = 0.030 / 0.047 / 0.065$ (N=8/16/20) · NV2 $\hat{\sigma} = 0.077$	Biased toward overestimation where dip signal mixes in = conservative upper bound
③ Noise mixing	Room-temp twin $\sigma = 0.06$ · cryo twin $\sigma = 0.02$ (white Gaussian) + T2 decay curves and contrast also randomized within measured ranges	Room temp matches the NV1 channel level; cryo follows literature conditions
④ Training randomization	Training data uses $\sigma \in [0.03, 0.12]$, uniformly random	Prevents the detector from specializing to one noise level (avoids overfitting)

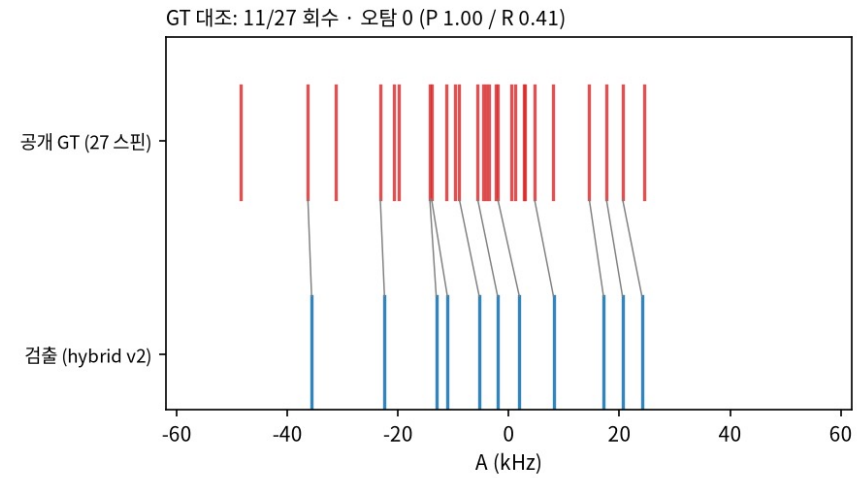
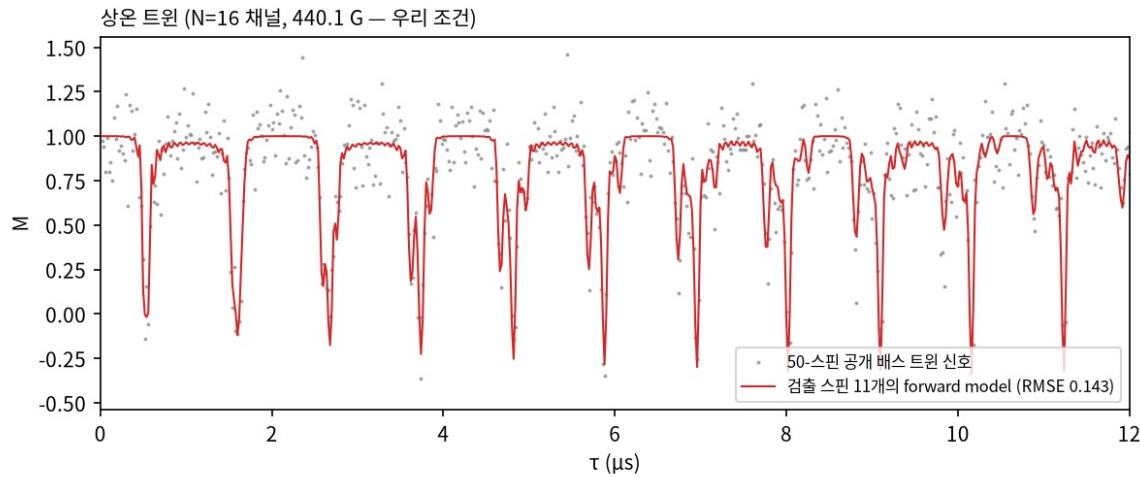
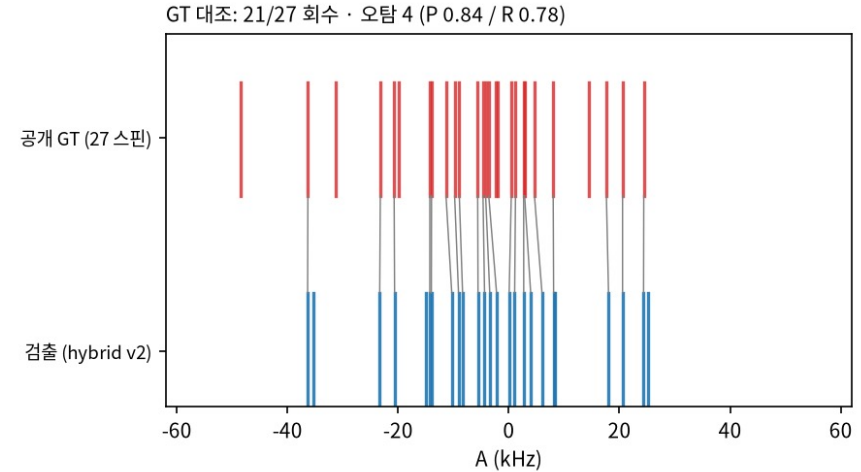
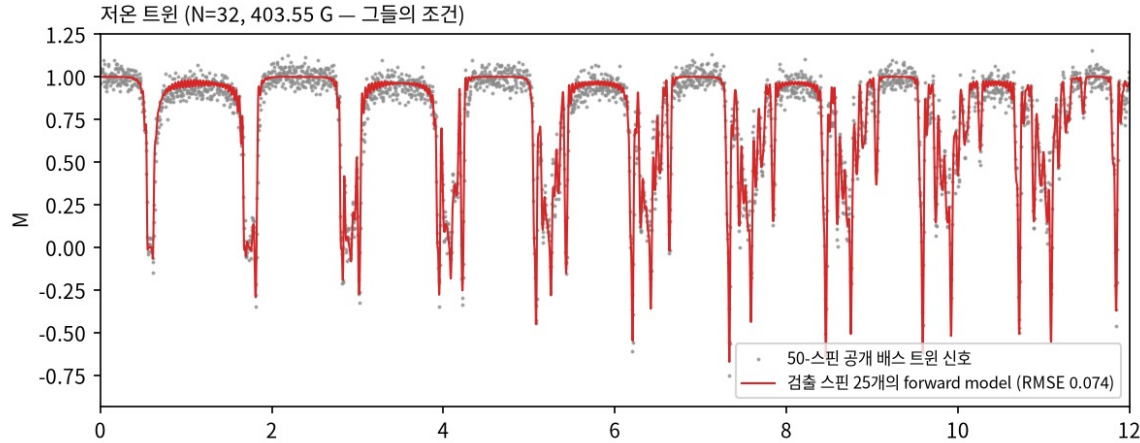


- Figure: zoom of raw NV1 (N=8/16/20) and NV2 (N=16) data (τ 4–7 μs) — scatter around the moving median (red) matches the $\pm 2\hat{\sigma}$ band
- Rightmost figure: per-channel $\hat{\sigma}$ vs adopted value (0.06) and training range (0.03–0.12)

Validation on the 50-Spin Data

Cryo data: 21/27 confirmed (RMSE 0.074) · room-temp mixed data: 11/27 · 0 false alarms — TEST spin values not used in training

공개 50-스핀 실측 배스(van de Stolpe 2024, 4TU)의 디지털 트윈 검증 — 신호는 공개 실측값으로 합성, 스핀 값은 학습에 미사용(테스트 전용)



• Benchmark summary (F1): the proposed hybrid v2 ranks 1st in every condition — cryo 0.84 / room temp 0.57 / error-injected 0.61 (2021 reproduction 0.38/0.37/0.35, classical DE 0.71/0.50/0.48, RJMCMC 0.84/0.56/0.52)

Final Confirmed NV1 / NV2 ¹³C Nuclear Spin Lists

NV1: 14 + NV2: 9 (+1 withheld) — spin lists confirmed by physics, statistics, and cross-validation

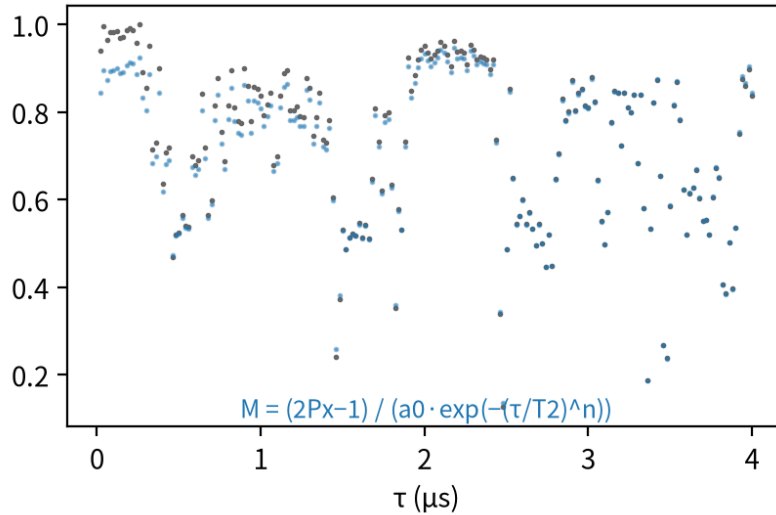
NV1 — 14 spins		NV2 — 9 spins + 1 withheld	
(−87.9 ±0.9, 18.3)	(+14.2 ±1.0, 21.7)	(+346.9, 261)	(−12.6, 42)
(−18.8 ±1.2, 15.5)	(+24.3 ±1.5, 20.1)	(−151.2, 95)	(−2.7, 37)
(−11.5 ±1.7, 13.9)	(+39.0 ±2.0, 28.2)	(−58.9, 182)	(+51.7, 91)
(−5.1 ±0.6, 22.5)	(+48.0 ±2.6, 25.1)	(−45.9, 50)	(+55.9, 55)
(+0.6 ±1.5, 17.1)	(+65.6 ±1.0, 27.6)	(−38.9, 67)	(+430.7, 58) withheld: unclear whether strongly coupled or degenerate
(+4.0 ±0.7, 26.2)	(+91.9 ±0.7, 32.6)		
(+8.4 ±0.6, 27.8)	(+114.4 ±1.7, 23.9)		

Count basis	BIC minimum k*=14 · consistent with RJMCMC posterior mode 15 (the most probable value in the count distribution) · only spins passing the addition penalty are accepted
Statistical reality	All 14 have mutually non-overlapping 95% CIs (60 bootstrap runs)
Physical reproduction	A single list explains all three N=8/16/20 measurements at once, RMSE 0.088/0.116/0.176 (reaching the noise floor)
External consistency	Confirmed consistent with manual-analysis anchors: NV1 4/4 · NV2 3/3
NV2 caveat	Single channel (CPMG-16), so A⊥ is a representative value · (+430.7) may be a degeneracy artifact → to be settled with additional CPMG-8/20 measurements
Off-list candidate	NV1 A ≈ −32±3 (supported by 3 series, not included in the final fit)

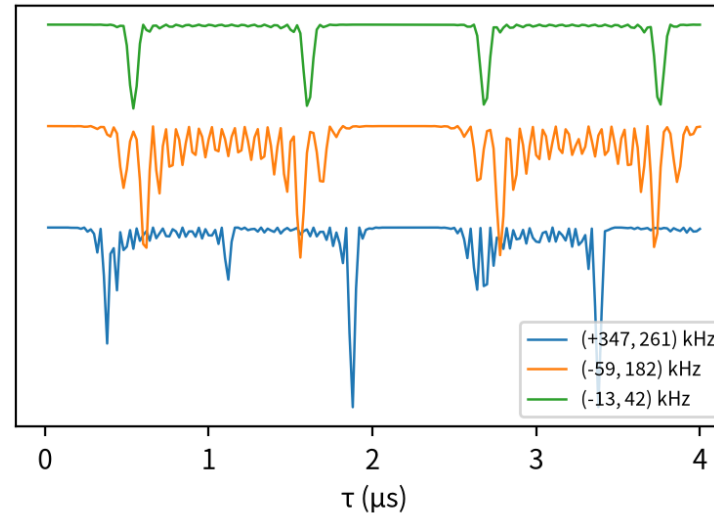
Fitting-Curve Synthesis — Product of Per-Spin Dips

How predicted values are computed from the spin list; the fitting-curve method

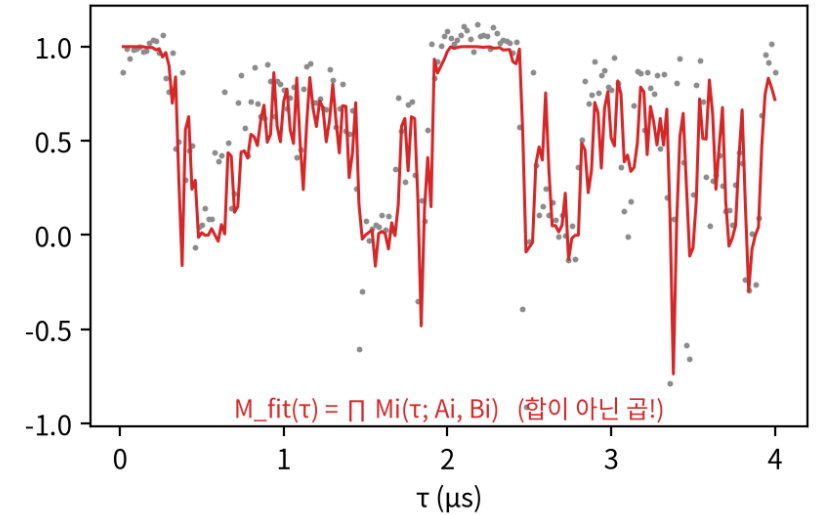
① 원시 $P_x \rightarrow$ 엔벨로프 정규화 M



② 스핀별 딥 트레인 $M_i(\tau)$



③ 곱 $\prod M_i =$ 피팅 곡선 (10스핀, RMSE 0.286)



Implemented per Jung 2021 Eq.1-3 · `cpmg/physics.py`

Eq.1 Single-spin dip: $M_k(\tau) = 1 - K \cdot \sin^2(N\varphi/2)$, $K = m_x^2 \cdot (1 - \cos \alpha)(1 - \cos \beta) / (1 + \cos \varphi)$

Eq.2 Phase: $\cos \varphi = \cos \alpha \cdot \cos \beta - m_z \cdot \sin \alpha \cdot \sin \beta$, $\alpha = \tilde{\omega} \tau$ (spin precession) · $\beta = \omega_L \tau$ (Larmor precession)

Eq.3 Spin axis: $\tilde{\omega} = \sqrt{(A_{\parallel} + \omega_L)^2 + A_{\perp}^2}$, $m_z = (A_{\parallel} + \omega_L) / \tilde{\omega}$, $m_x = A_{\perp} / \tilde{\omega}$

$\rightarrow$ Total product $M(\tau) = \prod_k M_k(\tau)$, $P_x = \frac{1}{2}(1 + a_0 \cdot M \cdot e^{-(\tau/T_2)^n})$

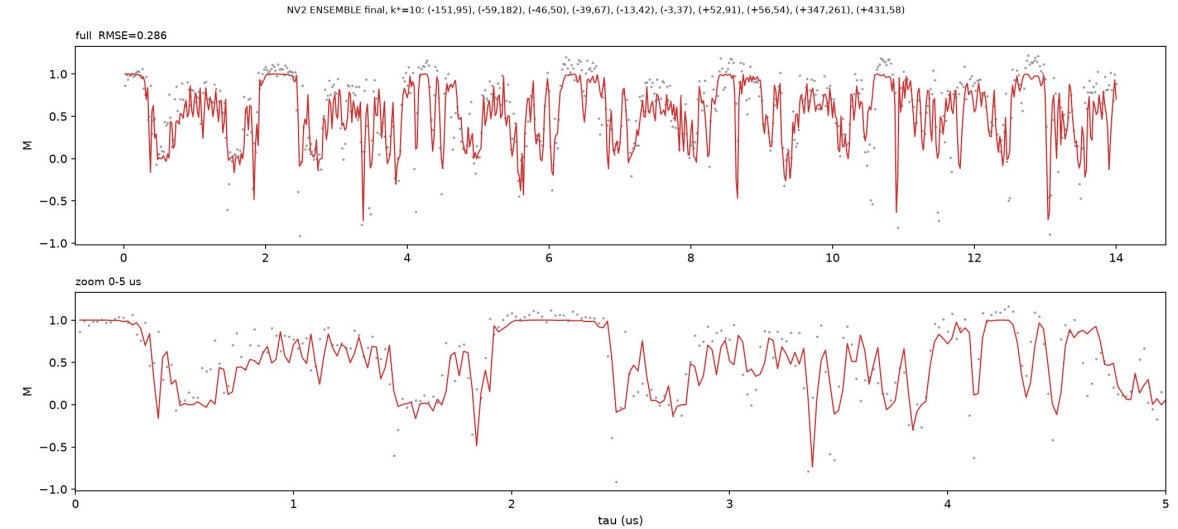
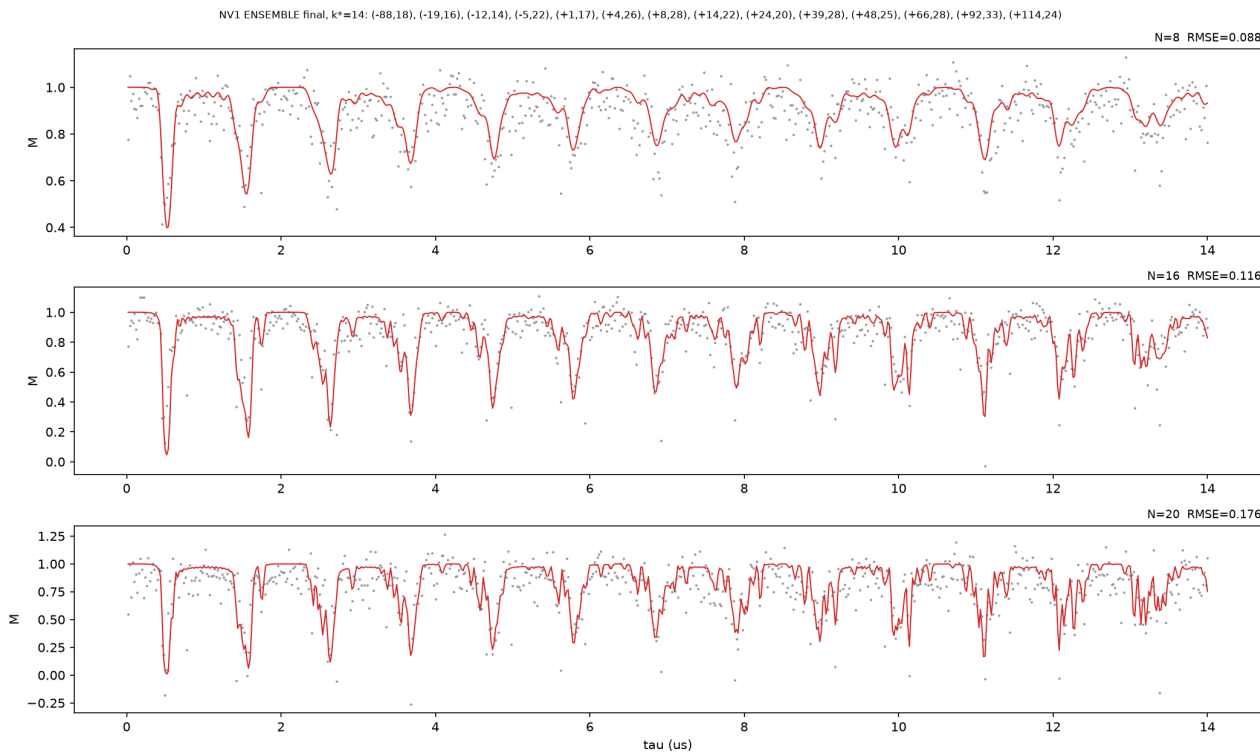
① Divide the experimental P_x by the envelope to normalize to the model's M scale $\rightarrow$ ② each spin (A_i, B_i) generates a dip train M_i at its own period $\rightarrow$ ③ the total product $\prod M_i$ is the fitting curve

Each added spin adds a factor to the product, reproducing finer structure

— On NV2, going from 6 to 10 spins lowers RMSE 0.308 $\rightarrow$ 0.286 (passes BIC; residual decrease = not overfitting)

Final Fit: Real CPMG Data (NV1: N=8/16/20 · NV2: N=16)

NV1: 14 spins reproduce all three measurements (N=8/16/20) simultaneously (RMSE 0.088/0.116/0.176) · NV2: 10 spins, RMSE 0.286



Gray = experiment · red = forward model from the detected spins

- Left, NV1: top→bottom N=8/16/20 — one and the same 14-spin list explains all three datasets
- Right, NV2: tracks even the strong-coupling fringe structure (1 spin withheld due to the single-channel limit)